Computer Networking: Beginner to Advanced Guide

Oct 12, 2026 · @Kaybee

How to use this guide

This guide takes you from "what is an IP address?" to designing and securing real networks in four stages. Work through them in order, and do the hands-on exercises at each stage, because networking is learned by doing.

Stage	You will learn	Time (part-time)
1. Beginner	Core concepts, devices, addressing, DNS, DHCP, OSI and TCP/IP models	2 to 4 weeks
2. Intermediate	Subnetting, switching, routing, VLANs, NAT, TCP vs UDP, troubleshooting	1 to 2 months
3. Advanced	OSPF, BGP, STP, QoS, IPv6, load balancing, SD-WAN	2 to 3 months
4. Security and cloud	Firewalls, VPNs, TLS, zero trust, cloud and virtual networking	2 to 3 months

Times are rough estimates and vary with how much you practise.

Stage 1: Networking basics

A network is two or more devices that can exchange data. The internet is a global network of networks. Everything below builds on a few core ideas.

Key devices

Device	What it does
Switch	Connects devices inside one network and forwards frames using MAC addresses

Device	What it does
Router	Connects different networks and forwards packets using IP addresses
Access point	Lets Wi-Fi devices join a wired network
Modem	Converts your ISP's signal (fibre, cable, DSL) into a form your router uses
Firewall	Allows or blocks traffic based on rules
NIC	The network card in your device, with a unique MAC address

At home, one box often combines the modem, router, switch, access point and firewall.

Addressing

- **MAC address**: a hardware address (for example `00:1A:2B:3C:4D:5E`) that works only inside the local network.
- **IPv4 address**: four numbers from 0 to 255, such as `192.168.1.25`. It identifies a device across networks.
- **Private ranges**: `10.0.0.0/8`, `172.16.0.0/12` and `192.168.0.0/16` are for internal use and are not routed on the internet.
- **Default gateway**: the router address your device sends traffic to when the destination is outside its own network.
- **Subnet mask**: tells a device which part of an IP address is the network and which part is the host (for example `255.255.255.0`, written `/24`).

Three services you use every day

- **DHCP** automatically gives devices an IP address, mask, gateway and DNS server when they connect.
- **DNS** translates names like `example.com` into IP addresses, like a phone book for the internet.
- **NAT** lets many private devices share one public IP address, which is how your home router works.

Common network types: LAN (one building), WAN (links between sites), WLAN (wireless LAN), and VPN (a private tunnel over the internet).

Try it now

1. On Windows, open Command Prompt and run `ipconfig /all`. On Mac or Linux, run `ifconfig` or `ip addr`.
2. Find your IP address, subnet mask, default gateway and DNS server.
3. Run `ping 8.8.8.8` (tests connectivity) and `ping google.com` (tests DNS as well). If the first works and the second fails, your problem is DNS.

Stage 1: The OSI and TCP/IP models

The OSI model splits networking into seven layers so you can reason about problems one layer at a time. Data moves down the layers on the sender and back up on the receiver, and each layer adds its own header (encapsulation).

Layer	Name	Job	Data unit	Examples
7	Application	Services apps use	Data	HTTP, DNS, SMTP, SSH
6	Presentation	Encoding, compression, encryption	Data	TLS, JPEG, ASCII
5	Session	Opening, managing and closing sessions	Data	NetBIOS, RPC
4	Transport	End-to-end delivery, ports, reliability	Segment / datagram	TCP, UDP
3	Network	Logical addressing and routing	Packet	IP, ICMP, OSPF
2	Data link	Local delivery using MAC addresses	Frame	Ethernet, Wi-Fi, ARP
1	Physical	Bits on the wire or air	Bits	Cables, fibre, radio

A classic memory aid from layer 7 down: "All People Seem To Need Data Processing."

TCP/IP model: the real internet uses a simpler four-layer model. Application (OSI 5 to 7), Transport (4), Internet (3), and Network Access (1 to 2).

How to use the layers when troubleshooting

- Cable unplugged or no Wi-Fi signal: layer 1.

- Wrong VLAN, MAC or switch problem: layer 2.
- Wrong IP, gateway or routing: layer 3.
- Port blocked or connection refused: layer 4.
- Website error, certificate or login problem: layers 5 to 7.

Stage 2: Intermediate concepts

Subnetting. Subnetting splits one network into smaller ones. The prefix length (the number after the slash) says how many bits are the network part. Hosts per subnet = 2^(32 − prefix) − 2.

Prefix	Mask	Usable hosts
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/30	255.255.255.252	2

Worked example: split `192.168.10.0/24` into four equal subnets. Borrow 2 bits to get /26, so each subnet has 64 addresses: `.0`, `.64`, `.128` and `.192`. The first address of each is the network address and the last is the broadcast address.

Switching and VLANs. A switch learns which MAC address sits behind which port and forwards frames only where needed. A VLAN splits one physical switch into several logical networks (for example Staff, Guests, Servers) to isolate traffic. Links between switches carry many VLANs using 802.1Q tagging and are called trunks.

Routing. A router chooses the next hop for each packet by looking up the destination in its routing table and picking the longest matching prefix. Routes come from three sources:

- **Connected**: networks directly attached to the router.
- **Static**: routes you configure by hand. Simple and predictable, but they do not adapt to failures.

- **Dynamic**: learned from protocols such as OSPF and BGP (see Stage 3).

NAT and PAT. NAT rewrites addresses as packets cross a router. PAT (the common form) also rewrites port numbers so thousands of private devices can share a single public IP.

TCP vs UDP

	TCP	UDP
Connection	Handshake first (SYN, SYN-ACK, ACK)	None
Reliability	Retransmits lost data, keeps order	None, "best effort"
Speed	Slower, more overhead	Faster, lighter
Used for	Web, email, file transfer, SSH	DNS queries, video calls, gaming, streaming

Common ports

Port	Protocol	Port	Protocol
22	SSH	80	HTTP
25	SMTP	443	HTTPS
53	DNS	67/68	DHCP
110 / 143	POP3 / IMAP	3389	Remote Desktop

ARP and ICMP. ARP finds the MAC address that belongs to an IP address on the local network. ICMP carries error and test messages, and is what `ping` and `traceroute` use.

Stage 2: Troubleshooting toolkit

Work from the bottom up: check the link, then the IP settings, then the gateway, then DNS, then the application. Each step tells you which layer to look at next.

Tool	Command (Windows / Linux or Mac)	Use it to
ping	`ping 8.8.8.8`	Test basic reachability and measure latency

Tool	Command (Windows / Linux or Mac)	Use it to
traceroute	`tracert host` / `traceroute host`	See each router hop and find where traffic stops
nslookup / dig	`nslookup example.com` / `dig example.com`	Check DNS resolution
ipconfig / ip	`ipconfig /all` / `ip addr`	Read your IP, mask, gateway and DNS
arp	`arp -a`	See the IP-to-MAC table
netstat / ss	`netstat -ano` / `ss -tulpn`	List open ports and active connections
curl	`curl -I https://example.com`	Test a web service and read its headers
Wireshark	GUI packet capture	Inspect real packets when nothing else explains the problem

A simple diagnostic ladder

1. `ping 127.0.0.1` works: your network stack is fine.
2. `ping <your gateway>` works: your local link is fine.
3. `ping 8.8.8.8` works: your internet route is fine.
4. `ping google.com` works: DNS is fine.
5. Everything pings but a service fails: check ports, firewalls and the application itself.

The first step that fails shows you where the fault is.

Stage 3: Advanced routing and switching

Dynamic routing protocols

Protocol	Type	Where it is used	Key idea
RIP	Distance vector	Tiny or legacy networks	Counts hops, max 15
OSPF	Link state	Inside one organisation (IGP)	Every router builds a map of the network and runs the shortest-path algorithm; uses areas to scale

Protocol	Type	Where it is used	Key idea
EIGRP	Advanced distance vector	Cisco-centric networks	Fast convergence, uses bandwidth and delay
BGP	Path vector	Between organisations and ISPs (EGP)	Exchanges routes between autonomous systems (AS) and picks paths by policy

BGP is the protocol that holds the internet together. It chooses routes by policy (attributes such as local preference, AS-path length and MED), not just by the shortest path. Misconfigured BGP can take whole networks offline, so changes need care.

Spanning Tree Protocol (STP). Redundant switch links create loops that can flood a network. STP blocks some links to give a loop-free tree and unblocks them if another link fails. Modern variants (RSTP, MSTP) converge in seconds instead of tens of seconds.

Link aggregation and redundancy. LACP bundles several physical links into one logical link for more bandwidth and failover. First-hop redundancy protocols (HSRP, VRRP) let two routers share one virtual gateway IP so a router failure is invisible to users.

Quality of Service (QoS). QoS decides which traffic gets priority when a link is congested. Voice and video are marked and queued ahead of bulk downloads so calls stay clear.

IPv6. IPv6 uses 128-bit addresses (for example `2001:db8::1`) and removes the need for NAT. Key points: a standard subnet is a /64, devices self-configure using SLAAC, and ARP is replaced by Neighbor Discovery. Most networks run IPv4 and IPv6 together (dual stack).

Load balancing and high availability. Load balancers spread traffic across servers and remove failed ones. Layer 4 balancers decide using IP and port. Layer 7 balancers can decide using the URL, headers or cookies.

SD-WAN and modern designs. SD-WAN manages many branch links (broadband, 4G/5G, MPLS) from one controller and steers traffic over the best path per application. Data-centre networks often use a spine-leaf design, where every leaf switch connects to every spine switch, for predictable low latency.

Network automation. At scale, engineers configure networks with code instead of by hand. Common tools are Python (Netmiko, NAPALM), Ansible, and APIs such as REST and NETCONF. Learn basic Python and Git alongside the protocols above.

Stage 4: Network security and cloud networking

Core security tools

Control	What it does
Firewall	Filters traffic by rules. Stateful firewalls track connections; next-generation firewalls also inspect applications and users
ACLs	Ordered permit/deny lists on routers and switches. The first match wins, with an implicit deny at the end
IDS / IPS	Detects (IDS) or detects and blocks (IPS) suspicious traffic
VPN	Encrypted tunnel. Site-to-site links offices; remote-access connects a user. Common types are IPsec, OpenVPN and WireGuard
TLS	Encrypts traffic between client and server (HTTPS) and proves the server's identity with certificates
802.1X / NAC	Requires devices and users to authenticate before getting network access
WPA3	Current Wi-Fi security standard

Good design principles

- **Segmentation**: separate guests, staff, servers and IoT devices into their own VLANs or subnets, with firewall rules between them.
- **Least privilege**: allow only the traffic that is needed and deny everything else.
- **Defence in depth**: layer several controls so one failure does not expose everything.
- **Zero trust**: never trust a device just because it is inside the network. Verify identity and device health for every request.
- **Logging and monitoring**: collect logs (syslog), flow data (NetFlow) and alerts (SNMP, SIEM) so you can see problems early.

Common attacks to understand (defensively): ARP spoofing, DNS poisoning, DDoS, man-in-the-middle, rogue access points and credential theft. Learn how each works so you can recognise and prevent it, and only test on networks you own or have written permission to test.

Cloud and virtual networking

- **VPC / VNet**: your private network inside AWS, Azure or Google Cloud, with subnets, route tables and gateways you define.
- **Security groups and network ACLs**: cloud firewalls at the instance and subnet level.
- **Hybrid connectivity**: site-to-site VPN or a dedicated link (AWS Direct Connect, Azure ExpressRoute) joins your office to the cloud.
- **Virtual networking**: hypervisors use virtual switches, and overlays such as VXLAN stretch layer 2 networks across a data centre.
- **Containers**: Kubernetes gives each pod an IP and uses services, ingress and network policies to control traffic.

Labs, certifications and resources

Free tools for hands-on practice

- **Cisco Packet Tracer**: free simulator for building routers, switches and VLANs (needs a free Cisco Networking Academy account).
- **GNS3 or EVE-NG**: run real router and switch images in virtual labs.
- **Wireshark**: capture and read your own traffic.
- **VirtualBox or VMware**: build small lab networks with virtual machines.
- **A Raspberry Pi or old PC**: run pfSense or OPNsense as a home firewall.

Projects, from easy to hard

1. Map your home network: list every device, its IP and MAC address.
2. Set a static IP and change your DNS server, then measure the difference.
3. Build a three-VLAN network (Staff, Guest, Servers) in Packet Tracer.
4. Subnet a /24 into four networks and make them route to each other.
5. Set up OSPF between three routers in GNS3.
6. Build a WireGuard VPN between your laptop and a home or cloud server.
7. Run a small DNS and DHCP server (for example with Pi-hole or dnsmasq).

Certifications

Level	Certification	Covers
Entry	CompTIA Network+	Vendor-neutral fundamentals
Entry to mid	Cisco CCNA	Routing, switching, IP services, security, automation
Security	CompTIA Security+	General security foundations
Advanced	Cisco CCNP, Juniper JNCIP	Enterprise and service-provider networking
Cloud	AWS Advanced Networking, Azure Network Engineer	Cloud networking design

Exam objectives and prices change, so check the vendor's website before you commit.

Study resources: Professor Messer's free Network+ course, Jeremy's IT Lab (free CCNA course on YouTube), the Cisco Networking Academy, and the book *Computer Networking: A Top-Down Approach* by Kurose and Ross for deeper theory.

These resources are listed from memory and were not verified online, so check that each is still available.

Part 2: Cisco Packet Tracer lab manual

This part turns the theory above into three labs you can build, test, document and put on a resume: a home network, a small office with VLANs, and a three-router hotel network running OSPF and SSH. Each lab has a topology, an addressing plan, the full CLI configuration and the commands that prove it works. Replace interface names and addresses with the ones in your own files if they differ.

Packet Tracer tech stack

Item	Recommended choice	Why
Simulator	Cisco Packet Tracer (current release, free via Cisco Networking Academy)	Runs real IOS-style CLI with no hardware
Router	Cisco 2911 or 2811 (ISR 4331 also works)	Has GigabitEthernet ports and slots for serial cards

Item	Recommended choice	Why
Serial card	HWIC-2T (add it with the router powered off)	Gives you Serial0/0/0 and Serial0/0/1 for the hotel lab
Switch	Cisco 2960 (Layer 2); 3560 or 3650 if you need a Layer 3 switch	Supports VLANs, trunks and port security
End devices	PC, Laptop, Printer, Server	Servers can run DHCP, DNS and HTTP
Wireless	Access Point-PT or a Wireless Router	For the home Wi-Fi extension
Documentation	Packet Tracer text boxes, screenshots, a `.pkt` file, a text file of configs	What you will submit as evidence

Cables to choose

Connection	Cable	Notes
PC to switch, switch to router	Copper straight-through	Different device types
Switch to switch, router to router (Ethernet), PC to router	Copper cross-over	Same device types (modern IOS ports often auto-sense, but use this in Packet Tracer to be safe)
Router to router (serial)	Serial DCE	Click the router first, then the other router. The side you click first is the DCE end and needs `clock rate`
PC console to device	Console (rollover)	Used to configure a device from a PC terminal

CLI modes

Prompt	Mode	Enter with	Leave with
`Router>`	User EXEC	Default	`exit`

Prompt	Mode	Enter with	Leave with
`Router#`	Privileged EXEC	`enable`	`disable`
`Router(config)#`	Global configuration	`configure terminal`	`end` or Ctrl+Z
`Router(config-if)#`	Interface configuration	`interface g0/0`	`exit`
`Router(config-line)#`	Line configuration	`line vty 0 4`	`exit`
`Router(config-router)#`	Routing protocol configuration	`router ospf 1`	`exit`

Universal commands you will use in every lab

```
enable
configure terminal
hostname HOTEL-R1
no ip domain-lookup
enable secret Cisco@123
service password-encryption
banner motd # Authorised access only #
line console 0
 password Cisco@console
 login
 logging synchronous
exit
copy running-config startup-config
```

Show commands to check your work

- `show running-config` : the active configuration.
- `show ip interface brief` : interface status and IP addresses at a glance (look for `up/up`).
- `show vlan brief` : VLANs and which ports belong to them.
- `show interfaces trunk` : trunk ports and allowed VLANs.
- `show ip route` : the routing table.
- `show ip ospf neighbor` : OSPF adjacencies.
- `show cdp neighbors` : which devices are directly connected.

Habits that save hours: label every device and interface in the workspace, save with `copy running-config startup-config` after every working step, build one section at a time and test it before moving on, and keep your configs in a text file as you go.

Project 1: Home network

Goal: one router and two switches serve two PCs and two printers, with automatic addressing (DHCP) and optional internet sharing (NAT). **Skills shown**: IP addressing, DHCP, static IPs for printers, switch management, NAT.

Topology (star network, with the second switch cascaded from the first)

```
                [ISP / Cloud]  (optional)
                      |  G0/1
                 [HOME-R1]
                      |  G0/0
                 [HOME-SW1]-----(F0/24)-----[HOME-SW2]
                  |      |                    |      |
                 PC1   Printer1             PC2   Printer2
```

Addressing plan

Device	Interface	Address	Method
HOME-R1	G0/0	192.168.1.1/24	Static (default gateway)
HOME-SW1	VLAN 1	192.168.1.2/24	Static (management)
HOME-SW2	VLAN 1	192.168.1.3/24	Static (management)
Printer1	NIC	192.168.1.10/24	Static
Printer2	NIC	192.168.1.11/24	Static
PC1, PC2	NIC	192.168.1.21 to .254	DHCP
HOME-R1	G0/1 (optional)	203.0.113.2/30	Static, toward the ISP

Router configuration

```
enable
configure terminal
hostname HOME-R1
no ip domain-lookup
enable secret Cisco@123
!
interface g0/0
 description LAN to HOME-SW1
 ip address 192.168.1.1 255.255.255.0
 no shutdown
!
ip dhcp excluded-address 192.168.1.1 192.168.1.20
ip dhcp pool HOME-LAN
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
!
end
copy running-config startup-config
```

The excluded range keeps `.1` to `.20` free for the router, switches and printers, so DHCP never hands those out.

Optional internet sharing (NAT overload), using a second router or cloud as the ISP:

```
configure terminal
interface g0/0
 ip nat inside
interface g0/1
 description To ISP
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 no shutdown
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
end
```

Switch configuration (repeat on SW2 with hostname `HOME-SW2` and address `192.168.1.3`)

```
enable
configure terminal
hostname HOME-SW1
no ip domain-lookup
enable secret Cisco@123
interface vlan 1
 ip address 192.168.1.2 255.255.255.0
 no shutdown
ip default-gateway 192.168.1.1
end
copy running-config startup-config
```

Static IPs on the printers: click the printer, open Config, then FastEthernet0, and set the IP address, subnet mask and gateway by hand. Set the PCs to DHCP under Desktop, then IP Configuration.

Verify

1. On a PC, run `ipconfig`: you should see a `192.168.1.x` address and gateway `192.168.1.1`.
2. `ping 192.168.1.1` from each PC, then ping the other PC and both printers.
3. On the router, run `show ip dhcp binding` to see the leases handed out.
4. With NAT: `ping 203.0.113.1` from a PC, then `show ip nat translations` on the router.

Wireless extension: add a Wireless Router or Access Point-PT to a switch port, set an SSID and WPA2 or WPA3 passphrase under Config, then connect a laptop with a wireless card.

Project 2: Small office with VLANs and inter-VLAN routing

Goal: separate two departments into their own VLANs on one switch, then let them talk to each other through the router. **Skills shown**: VLAN creation, access and trunk ports, router-on-a-stick (subinterfaces with 802.1Q), per-VLAN DHCP, native and management VLAN.

Topology (star, with one trunk link carrying both VLANs to the router)

```
   PC1 (Sales)  PC2 (Sales)  PC3 (Admin)  PC4 (Admin)
     F0/1         F0/2         F0/3         F0/4
       \            |            |            /
        +---------[ OFFICE-SW1 ]-------------+
                       | G0/1  (trunk: VLANs 10, 20, 99)
                       | G0/0.10  G0/0.20  G0/0.99
                  [ OFFICE-R1 ]
```

VLAN and addressing plan

VLAN	Name	Subnet	Gateway (router subinterface)	Switch ports
10	SALES	192.168.10.0/24	192.168.10.1 (G0/0.10)	F0/1 to F0/2
20	ADMIN	192.168.20.0/24	192.168.20.1 (G0/0.20)	F0/3 to F0/4
99	MGMT (native)	192.168.99.0/24	192.168.99.1 (G0/0.99)	Trunk only, switch IP .2

Switch configuration

```
enable
configure terminal
hostname OFFICE-SW1
no ip domain-lookup
vlan 10
 name SALES
vlan 20
 name ADMIN
vlan 99
 name MGMT
!
interface range f0/1 - 2
 switchport mode access
 switchport access vlan 10
interface range f0/3 - 4
 switchport mode access
 switchport access vlan 20
!
interface g0/1
 description Trunk to OFFICE-R1
```

```
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,99
!
interface vlan 99
 ip address 192.168.99.2 255.255.255.0
 no shutdown
ip default-gateway 192.168.99.1
end
copy running-config startup-config
```

Router configuration

```
enable
configure terminal
hostname OFFICE-R1
no ip domain-lookup
interface g0/0
 no shutdown
!
interface g0/0.10
 description SALES gateway
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface g0/0.20
 description ADMIN gateway
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface g0/0.99
 description MGMT and native VLAN
 encapsulation dot1Q 99 native
 ip address 192.168.99.1 255.255.255.0
!
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp pool SALES
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool ADMIN
 network 192.168.20.0 255.255.255.0
```

```
 default-router 192.168.20.1
 dns-server 8.8.8.8
end
copy running-config startup-config
```

The physical interface `g0/0` has no IP address. Only the subinterfaces do. The number after the dot is just a label, but matching it to the VLAN ID makes the config easier to read.

Verify

1. `show vlan brief` on the switch: VLAN 10 holds F0/1 and F0/2, VLAN 20 holds F0/3 and F0/4.
2. `show interfaces trunk`: G0/1 is trunking with VLANs 10, 20 and 99 allowed.
3. `show ip interface brief` on the router: all three subinterfaces `up/up`.
4. PC1 `ping 192.168.10.1`, then `ping` a PC in VLAN 20. A successful cross-VLAN ping proves inter-VLAN routing works.
5. `tracert 192.168.20.x` from a Sales PC: the first hop should be `192.168.10.1`, the router.

Common mistakes: forgetting `no shutdown` on the physical interface, setting the switch port to access instead of trunk, a VLAN missing from the allowed list, a native VLAN mismatch between switch and router, and PCs left on static addresses instead of DHCP.

Variation: use a Layer 3 switch (3560) instead of router-on-a-stick. Run `ip routing`, then create `interface vlan 10` and `interface vlan 20` with the gateway addresses. This is faster and how larger networks do it.

Project 3: Hotel management network (3 routers, OSPF, SSH)

Goal: one router per floor, linked by serial WAN-style links, running OSPF so every floor reaches every other, with all routers managed over SSH from the IT department PC. **Skills shown**: serial DCE/DTE links and `clock rate`, /30 point-to-point addressing, OSPF, router hardening, remote management with SSH.

Topology (a chain: Floor 1, Floor 2, Floor 3 and IT)

```
[Floor 1 LAN]--[HOTEL-R1]==S0/0/0 (DCE) ---- S0/0/0==[HOTEL-R2]==S0/0/1 (DCE)
---- S0/0/0==[HOTEL-R3]--[Floor 3 / IT LAN]
  192.168.1.0/24            10.0.12.0/30                        10.0.23.0/30
192.168.3.0/24
                                                 |
                                            [Floor 2 LAN]
                                            192.168.2.0/24
```

Each LAN has a switch and its end devices behind the router's `G0/0`. The IT PC (`192.168.3.10`) sits on Floor 3 behind HOTEL-R3.

Addressing plan

Device	Interface	Address	Notes
HOTEL-R1	G0/0	192.168.1.1/24	Floor 1 LAN gateway
HOTEL-R1	S0/0/0	10.0.12.1/30	**DCE**, needs clock rate
HOTEL-R2	S0/0/0	10.0.12.2/30	DTE side of the R1 link
HOTEL-R2	S0/0/1	10.0.23.1/30	**DCE**, needs clock rate
HOTEL-R2	G0/0	192.168.2.1/24	Floor 2 LAN gateway
HOTEL-R3	S0/0/0	10.0.23.2/30	DTE side of the R2 link
HOTEL-R3	G0/0	192.168.3.1/24	Floor 3 and IT gateway
IT PC	NIC	192.168.3.10/24	Gateway 192.168.3.1

In Packet Tracer the DCE end is the router you clicked first when drawing the serial cable. A small clock icon shows on the DCE side. Confirm it with `show controllers serial 0/0/0`.

Step 1: HOTEL-R1 (interfaces, OSPF, SSH)

```
enable
configure terminal
hostname HOTEL-R1
no ip domain-lookup
enable secret Cisco@123
service password-encryption
banner motd # Authorised access only #
!
```

```
interface g0/0
 description Floor 1 LAN
 ip address 192.168.1.1 255.255.255.0
 no shutdown
interface s0/0/0
 description Link to HOTEL-R2
 ip address 10.0.12.1 255.255.255.252
 clock rate 64000
 no shutdown
!
router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
 passive-interface g0/0
!
ip domain-name hotel.local
username admin secret Admin@123
crypto key generate rsa modulus 2048
ip ssh version 2
access-list 10 permit host 192.168.3.10
line vty 0 4
 login local
 transport input ssh
 access-class 10 in
 exec-timeout 5 0
end
copy running-config startup-config
```

Step 2: HOTEL-R2 (DTE toward R1, DCE toward R3)

```
enable
configure terminal
hostname HOTEL-R2
no ip domain-lookup
enable secret Cisco@123
service password-encryption
banner motd # Authorised access only #
!
interface g0/0
 description Floor 2 LAN
 ip address 192.168.2.1 255.255.255.0
 no shutdown
```

```
interface s0/0/0
 description Link to HOTEL-R1
 ip address 10.0.12.2 255.255.255.252
 no shutdown
interface s0/0/1
 description Link to HOTEL-R3
 ip address 10.0.23.1 255.255.255.252
 clock rate 64000
 no shutdown
!
router ospf 1
 router-id 2.2.2.2
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.23.0 0.0.0.3 area 0
 passive-interface g0/0
!
ip domain-name hotel.local
username admin secret Admin@123
crypto key generate rsa modulus 2048
ip ssh version 2
access-list 10 permit host 192.168.3.10
line vty 0 4
 login local
 transport input ssh
 access-class 10 in
 exec-timeout 5 0
end
copy running-config startup-config
```

Step 3: HOTEL-R3 (Floor 3 and IT)

```
enable
configure terminal
hostname HOTEL-R3
no ip domain-lookup
enable secret Cisco@123
service password-encryption
banner motd # Authorised access only #
!
interface g0/0
 description Floor 3 and IT LAN
 ip address 192.168.3.1 255.255.255.0
```

```
 no shutdown
interface s0/0/0
 description Link to HOTEL-R2
 ip address 10.0.23.2 255.255.255.252
 no shutdown
!
router ospf 1
 router-id 3.3.3.3
 network 192.168.3.0 0.0.0.255 area 0
 network 10.0.23.0 0.0.0.3 area 0
 passive-interface g0/0
!
ip domain-name hotel.local
username admin secret Admin@123
crypto key generate rsa modulus 2048
ip ssh version 2
access-list 10 permit host 192.168.3.10
line vty 0 4
 login local
 transport input ssh
 access-class 10 in
 exec-timeout 5 0
end
copy running-config startup-config
```

The `network` statements use wildcard masks (the inverse of a subnet mask): `0.0.0.255` for a /24 and `0.0.0.3` for a /30. `passive-interface g0/0` stops OSPF hellos going out to PCs while still advertising the LAN.

Step 4: Test from the IT PC

Open Desktop, then Command Prompt on the IT PC:

```
ping 192.168.1.1
ping 192.168.2.1
ssh -l admin 10.0.12.1
ssh -l admin 10.0.23.1
ssh -l admin 192.168.3.1
```

Enter the password when asked. A router prompt such as `HOTEL-R1#` after login proves remote management works. Using the loopback or serial addresses all work, because OSPF gives the IT PC a route to each one.

Verify

Command	Healthy result
`show ip interface brief`	Serial and Gigabit interfaces `up/up`
`show controllers serial 0/0/0`	Shows DCE cable on the correct side
`show ip ospf neighbor`	Neighbours in state `FULL` (R2 sees two)
`show ip route ospf`	Routes marked `O` to the other floors' LANs
`show ip protocols`	OSPF process 1 with the right networks and router ID
`show ssh`	Active SSH sessions after you connect

Troubleshooting this lab

- Serial shows `up/down`: the DCE side is missing `clock rate`, or the encapsulation or subnet does not match.
- No OSPF neighbour: check both ends are in the same /30, the same area, and neither is passive.
- SSH refused: the `crypto key generate rsa` step was skipped, the domain name was not set before it, or the IT PC address is missing from access list 10.
- Password fails: SSH uses the `username` account, not the line password.

Which topology fits which protocol

A topology is the shape of the connections. Pick it by asking what must stay up, how much traffic grows, and what you can afford.

Physical and logical topologies

Topology	Shape	Strength	Weakness	Typical use
Star	Everything connects to a central switch	Easy to manage, one failure is isolated	Central device is a single point of failure	Homes, offices, most LANs

Topology	Shape	Strength	Weakness	Typical use
Extended star / hierarchical	Stars linked to a core	Scales well, clear layers	Needs planning	Campuses, hotels, large offices
Ring	Each device links to two neighbours	Predictable paths	One break can hurt without redundancy	Metro fibre rings, legacy token ring
Full mesh	Every device links to every other	Highest redundancy	Cost and links grow fast: n(n−1)/2	Data-centre cores, critical WAN sites
Partial mesh	Some extra redundant links	Balanced cost and resilience	More complex routing	Enterprise WANs
Hub-and-spoke	Branches connect to one central site	Cheap, simple	Hub failure isolates everyone	Branch offices, VPNs
Point-to-point	Two devices, one link	Simple, dedicated	Does not scale	Serial and leased lines (your hotel links)
Bus	One shared cable	Cheap (obsolete)	Collisions, one break stops all	Old Ethernet; exam history only

Protocol and technology to topology

Protocol or technology	Best topology	Why and what it is used for	Packet Tracer lab
Static routing	Small stub or hub-and-spoke	Simple, no protocol overhead, but you update it by hand	Two routers and a default route
RIP	Small, flat networks	Easy to learn, slow, limited to 15 hops	3 routers in a line

Protocol or technology	Best topology	Why and what it is used for	Packet Tracer lab
OSPF	Hierarchical, partial or full mesh, multi-area	Fast convergence and a full map of the network; the standard inside an organisation	Your hotel lab, then add a fourth router and a second area
EIGRP	Same as OSPF in Cisco-only networks	Fast, easy to configure	3 routers in a triangle
BGP	Between separate organisations (ISP edges)	Policy-based routing between autonomous systems	2 or 3 routers, each in its own AS
VLANs	Star around a switch	Splits one switch into several logical networks	Your small office lab
Inter-VLAN routing	Router with a trunk (router-on-a-stick) or a Layer 3 switch	Lets VLANs communicate under control	Your small office lab
STP and RSTP	Switches in a triangle or ring with redundant links	Stops loops while keeping a backup path	3 switches in a triangle, then pull a cable
EtherChannel (LACP)	Two switches joined by 2 to 8 parallel links	More bandwidth and failover on one logical link	2 switches with 2 links
HSRP / VRRP	Two routers on the same LAN	One virtual gateway that survives a router failure	2 routers plus 1 switch
DHCP	Centralised server, or relay across routers	Automatic addressing; use `ip helper-address` when the server is on another subnet	Server on one LAN, clients on another
NAT / PAT	Network edge, facing the ISP	Shares one public IP among many private hosts	Home lab with an ISP router

Protocol or technology	Best topology	Why and what it is used for	Packet Tracer lab
ACLs	Between zones, near the source (extended) or near the destination (standard)	Controls who may reach what	Block guests from servers
SSH / AAA	Management VLAN or an IT PC, as in your hotel lab	Secure remote administration	Your hotel lab
Site-to-site VPN (IPsec, GRE)	Hub-and-spoke or point-to-point over a WAN or the internet	Joins offices over a public network securely	2 routers with a cloud in between
Wireless (WLAN)	Access points on the access layer, optionally with a controller	Mobility for staff and guests	AP on a switch, laptop joins

The three-tier hierarchical design is the pattern most real networks follow, and Network+ tests it:

- **Access layer**: switches that end devices plug into. VLANs, port security and PoE live here.
- **Distribution layer**: aggregates access switches and applies policy. Inter-VLAN routing, ACLs and summarisation live here.
- **Core layer**: fast backbone. Keep it simple and redundant, and avoid putting policy here.

Small networks use a **collapsed core**, where distribution and core are one device pair. Modern data centres use **spine-leaf**.

Rule of thumb: use the simplest design that meets the availability need. Add redundancy (a second link, a second router) only for the parts whose failure would hurt the business.

Building secure networks in Packet Tracer

Security is built in layers: secure the devices, secure the management access, secure the traffic between zones, and secure the routing. Apply these to your three labs. Use your own passwords, and never reuse the lab ones on real equipment.

1. Secure the router

```
enable secret Str0ng!Enable
service password-encryption
security passwords min-length 10
banner motd # Authorised access only. Activity is logged. #
login block-for 120 attempts 3 within 60
line console 0
 password C0nsole!Pass
 login
 exec-timeout 5 0
 logging synchronous
line aux 0
 no exec
```

Use `enable secret` (hashed) rather than `enable password`. `login block-for` slows down password-guessing by locking logins for 120 seconds after 3 failures in 60 seconds. Note that `service password-encryption` only lightly obscures passwords. It is not strong encryption.

2. Secure remote access (SSH only, from one trusted PC)

```
ip domain-name hotel.local
username admin secret Adm1n!Pass
crypto key generate rsa modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
access-list 10 permit host 192.168.3.10
line vty 0 4
 login local
 transport input ssh
 access-class 10 in
 exec-timeout 5 0
```

Telnet sends passwords in clear text, which is why `transport input ssh` matters. The standard ACL limits management to the IT PC.

3. Secure the switch

```
! Shut down and park unused ports
interface range f0/5 - 24
 switchport mode access
 switchport access vlan 999
 shutdown
!
! Port security on access ports
interface range f0/1 - 4
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation restrict
!
! Stop rogue switches and loops on access ports
interface range f0/1 - 4
 spanning-tree portfast
 spanning-tree bpduguard enable
!
! Avoid VLAN hopping
interface g0/1
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,99
 switchport nonegotiate
```

Create VLAN 999 first (`vlan 999`, `name PARKING`). Check with `show port-security interface f0/1` and `show interfaces status`. Use `violation shutdown` instead of `restrict` when you want a strict response, but it disables the port until you reset it.

4. Stop rogue DHCP servers (DHCP snooping)

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
interface g0/1
 ip dhcp snooping trust
```

Only the trusted uplink toward the real DHCP server is allowed to answer DHCP requests. Support for snooping and Dynamic ARP Inspection varies by switch model and Packet Tracer version, so check your model.

5. Control traffic between zones with ACLs (small office)

Example policy: Sales may use the Admin web server but nobody else may reach it; Sales may not reach other Admin hosts.

```
ip access-list extended SALES-TO-ADMIN
 permit tcp 192.168.10.0 0.0.0.255 host 192.168.20.50 eq 80
 permit tcp 192.168.10.0 0.0.0.255 host 192.168.20.50 eq 443
 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 permit ip any any
interface g0/0.10
 ip access-group SALES-TO-ADMIN in
```

ACL rules are read top to bottom and the first match wins. Every ACL ends with an invisible `deny any`. Place **extended** ACLs close to the source and **standard** ACLs close to the destination. Verify with `show access-lists` and test with pings and web requests.

6. Secure OSPF (hotel network)

```
interface s0/0/0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 0spfKey!
router ospf 1
 passive-interface g0/0
```

Configure the same key on both ends of each link. Authentication stops a rogue router from injecting false routes. Passive interfaces stop OSPF from talking to end users. In real networks, prefer SHA-based OSPF authentication where supported.

7. Segment and isolate

- Put guests, staff, servers, printers and management in separate VLANs.
- Give guests internet only (an ACL denying private ranges).
- Hotel guests and the hotel management systems should never share a subnet.
- Keep the management VLAN off user ports.

8. Log and monitor

```
service timestamps log datetime msec
logging host 192.168.3.20
logging trap informational
ntp server 192.168.3.20
snmp-server community Read0nly!Str ro 10
```

A Packet Tracer server can act as a syslog, NTP and AAA server. Real incidents are only traceable when device clocks agree.

Security checklist before you call a lab finished

- [ ] No default or shared passwords, and all secrets hashed
- [ ] Telnet disabled, SSH v2 only
- [ ] Management restricted to one host or VLAN
- [ ] Unused ports shut down and in a parking VLAN
- [ ] Port security on access ports
- [ ] Native VLAN is not VLAN 1 and trunks allow only needed VLANs
- [ ] ACLs enforce the policy you wrote down
- [ ] Routing protocol authentication on
- [ ] Configs saved and a copy stored outside the device

Documenting your projects and putting them on a resume

Employers cannot see your Packet Tracer file, so the documentation is the proof. Aim for one clear page per project plus the files behind it.

Project write-up template (one per lab)

Section	What to include
Title and purpose	One sentence: the business problem the network solves
Requirements	What it had to do (for example, "Sales and Admin isolated but able to reach shared printers")
Topology diagram	A clean screenshot with device names and interface labels
Addressing table	Devices, interfaces, IPs, masks, gateways (as in the labs above)
VLAN table	VLAN IDs, names, subnets, ports
Technologies used	VLANs, 802.1Q trunking, OSPF, DHCP, NAT, SSH, ACLs and so on
Configuration	The key CLI blocks, with comments explaining why

Section	What to include
Verification	Screenshots of `show ip route`, `show ip ospf neighbor`, pings and SSH logins
Problems and fixes	What broke and how you diagnosed it (this impresses interviewers most)
Security measures	What you hardened and why
What I learned / next steps	One or two honest lines

Where to keep it: create a public GitHub repository (for example `networking-labs`) with one folder per project holding the `.pkt` file, a `README.md` with the write-up, a `configs/` folder of `.txt` files for each device, and a `screenshots/` folder. Put the link on your resume and LinkedIn.

Resume bullets you can adapt. Only keep what you actually built and tested, and replace anything in brackets with your real detail.

Section heading: Networking Projects (Cisco Packet Tracer lab projects)

- **Hotel network (3 sites)**: Designed and configured a three-router hotel network, one router per floor, linked by serial DCE connections with clock rates set on the DCE interfaces, using point-to-point /30 subnets. Implemented OSPF so all floors could reach each other, and verified adjacencies and routes with `show ip ospf neighbor` and `show ip route`.
- **Remote management**: Configured SSH version 2 on all routers so the IT department PC could manage every router remotely, replacing Telnet, with local user accounts and an access list limiting access to the IT host.
- **Small office with VLANs**: Built a small office network with two VLANs on one switch, an 802.1Q trunk to a router, and inter-VLAN routing using subinterfaces (router-on-a-stick) with DHCP per VLAN, and verified connectivity between four end devices.
- **Home network**: Built a home network with one router, two switches, two PCs and two printers, with DHCP, static addressing for printers and [NAT for internet sharing if you added it].
- **Documentation**: Documented each lab with topology diagrams, IP addressing tables, configuration files and test evidence, published on [GitHub link].

A skills line for your resume: Cisco IOS CLI, VLANs, 802.1Q trunking, inter-VLAN routing, OSPF, static routing, DHCP, NAT, SSH, subnetting (IPv4), network documentation, Packet Tracer, Wireshark (if you used it).

Honesty rules (they protect you in interviews)

- Label these as lab or simulated projects. Employers value them as practice when they are described plainly.
- Do not claim real-world production experience or numbers you did not measure.
- Do not write "CCNA" or "Network+" until you hold the certification. "Studying for CompTIA Network+, exam planned [month, year]" is honest and shows direction.

Be ready to answer these (practise out loud)

1. Why did you use /30 subnets on the serial links?
2. What does `clock rate` do and which end needs it?
3. How does OSPF choose the best path, and what is a router ID?
4. What is the difference between an access port and a trunk port?
5. How does router-on-a-stick work, and what are its limits?
6. Why is SSH better than Telnet?
7. Walk me through how you would troubleshoot a PC that cannot reach another VLAN.

If you can answer those clearly, you understand your own projects, which is exactly what an interviewer is testing.

Upgrades for each project and a troubleshooting checklist

Once a lab works, upgrading it is the fastest way to learn and gives you a stronger story on your resume.

Project	Upgrade	New skill it proves
Home network	Add Wi-Fi with WPA2/WPA3, a guest network, NAT, and a DNS and web server	Wireless, NAT, server services
Home network	Connect a second router and run static and default routes	Static routing
Small office	Add VLAN 30 (Guests) with an ACL that allows internet only	Segmentation and ACLs

Project	Upgrade	New skill it proves
Small office	Add a second switch with a trunk and a redundant link	STP and EtherChannel
Small office	Replace router-on-a-stick with a Layer 3 switch	SVI routing
Hotel	Add a DHCP server or relay (`ip helper-address`)	DHCP across routers
Hotel	Add a second link from R1 to R3 and watch OSPF reroute	Redundancy and convergence
Hotel	Add loopback addresses, OSPF authentication and `ip ospf cost` tuning	Hardening and path control
Hotel	Add a branch router joined by an IPsec or GRE tunnel	VPN
Hotel	Add a guest Wi-Fi VLAN, a syslog and NTP server	Monitoring and wireless

Troubleshooting checklist

When something fails, move from the bottom layer up and change one thing at a time.

1. **Is the link up?** `show ip interface brief`. `down/down` means cable or port; `up/down` usually means a mismatch (clock rate, encapsulation, VLAN or trunk).
2. **Is the IP configuration right?** Check address, mask and gateway on the PC with `ipconfig`.
3. **Can it reach the gateway?** `ping` the gateway first, then a host beyond it.
4. **Does the router have a route?** `show ip route` on every router along the path. Routing must work in both directions.
5. **Is something blocking it?** `show access-lists`, port security status, and VLAN or trunk configuration.
6. **Is name resolution working?** Ping an IP, then a name.
7. **Compare with the working lab**: `show running-config` and look for typos in VLAN IDs, masks and `network` statements.

Symptom to likely cause

Symptom	Likely cause
Serial `up/down`	Missing `clock rate` on DCE, or mismatched encapsulation
PC gets `169.254.x.x`	DHCP not reachable, pool wrong or excluded range too large
Cross-VLAN ping fails	Trunk down, VLAN not allowed on trunk, missing subinterface or wrong gateway
OSPF neighbour stuck in `INIT` or `2WAY`	Mismatched hello timers, authentication or area, or one side passive
SSH "connection refused"	No RSA key, no domain name, no `transport input ssh`, or ACL blocking
First ping drops, rest succeed	Normal: ARP resolution on the first packet
Ping works one way only	Missing return route or an ACL on one side

Part 3: CompTIA Network+ study plan and methodology

Network+ rewards people who can reason through a scenario, not people who only memorise. You already have hands-on labs, so the plan below builds the theory and exam technique around what you can do.

Exam facts (as published for version N10-009)

- Up to 90 questions in 90 minutes, mixing multiple-choice and performance-based questions (PBQs), where you configure or troubleshoot a simulated setup.
- Passing score: 720 on a scale of 100 to 900.
- CompTIA recommends A+ and 9 to 12 months of networking experience, but neither is required.
- CompTIA releases new versions every few years. Before you book in 2027, confirm on the official CompTIA exam page that N10-009 is still the current version and download its free objectives PDF. Study from that list.

The five domains and where to spend your time

Domain	Weight	Core topics to master first	Covered in this guide
1. Networking Concepts	23%	OSI and TCP/IP models, ports and protocols, IPv4 addressing and subnetting, IPv6 basics, topologies, cables and connectors, Ethernet standards, cloud concepts	Stages 1 and 2, topology section
2. Network Implementation	20%	Routing (static, OSPF, BGP basics), switching (VLANs, trunks, STP, link aggregation), wireless standards and design, device placement	Stage 3, Projects 2 and 3
3. Network Operations	19%	Documentation, change management, monitoring (SNMP, syslog, NetFlow), high availability and disaster recovery, DHCP, DNS, NTP, VPN and remote access	Stage 4, documentation section
4. Network Security	14%	CIA triad, defence in depth, zero trust, attacks and mitigations, ACLs, port security, 802.1X, AAA, hardening	Security section, Stage 4
5. Network Troubleshooting	24%	The troubleshooting methodology, cable and connectivity faults, DNS and DHCP faults, performance problems, command-line tools	Troubleshooting sections

Troubleshooting and Networking Concepts together are almost half the exam, so they get the most time. Network Security is the smallest domain, but its topics reappear inside scenario questions elsewhere.

The troubleshooting methodology (memorise this)

CompTIA tests a fixed six-step process, and questions often ask "what should be done next?" Always pick the step that follows the one already completed.

1. **Identify the problem.** Gather information, question the user, identify symptoms, determine what changed, duplicate the problem if possible, approach multiple problems one at a time.
2. **Establish a theory of probable cause.** Question the obvious and consider multiple approaches (top-to-bottom or bottom-to-top OSI, divide and conquer).
3. **Test the theory to determine the cause.** If the theory is confirmed, go to step 4. If not, establish a new theory or escalate.
4. **Establish a plan of action to resolve the problem and identify potential effects, then implement the solution.**
5. **Verify full system functionality and implement preventive measures.**
6. **Document findings, actions, outcomes and lessons learned.**

Example: users on VLAN 20 cannot reach the file server. Step 1: ask when it started and what changed, and find that only VLAN 20 is affected. Step 2: theory that the trunk does not allow VLAN 20. Step 3: run `show interfaces trunk` and see VLAN 20 missing. Step 4: add the VLAN to the allowed list (check it affects nothing else). Step 5: ping from VLAN 20 and ask the users to confirm. Step 6: record the cause and fix.

Must-know-cold list

These appear on almost every exam and in daily work. Make flashcards for them.

- **Ports**: 20/21 FTP, 22 SSH/SFTP, 23 Telnet, 25 SMTP, 53 DNS, 67/68 DHCP, 69 TFTP, 80 HTTP, 110 POP3, 123 NTP, 143 IMAP, 161/162 SNMP, 389 LDAP, 443 HTTPS, 445 SMB, 514 syslog, 636 LDAPS, 1433 SQL Server, 3306 MySQL, 3389 RDP.
- **OSI layers** and what operates at each (devices, protocols, data units).
- **Subnetting**: convert a /prefix to a mask, find hosts per subnet, network and broadcast addresses, and split a block into equal subnets. Practise ten problems a day until it takes seconds.
- **Private ranges and special addresses**: RFC 1918 ranges, APIPA `169.254.0.0/16`, loopback `127.0.0.1`.
- **Cables and standards**: Cat 5e, 6, 6a, 8; multimode and single-mode fibre; common connectors (RJ45, LC, SC, ST, MPO); T568A and T568B wiring; Ethernet standards such as 1000BASE-T and 10GBASE-SR; maximum distances (100 m for copper).

- **Wireless**: 802.11 standards (a/b/g/n/ac/ax), 2.4 GHz vs 5 GHz vs 6 GHz, channels 1/6/11, WPA2 vs WPA3, SSID, roaming.
- **DNS records**: A, AAAA, CNAME, MX, NS, PTR, TXT, SRV.
- **DHCP**: the DORA process (Discover, Offer, Request, Acknowledge), scopes, reservations, relay.
- **Routing and switching**: administrative distance, OSPF vs BGP vs EIGRP, STP port states and roles, VLAN and trunk behaviour, EtherChannel, PoE.
- **Security**: CIA triad, common attacks (DoS/DDoS, spoofing, ARP poisoning, VLAN hopping, rogue DHCP, evil twin, phishing), mitigations, AAA, RADIUS vs TACACS+, NAC and 802.1X, VPN types.
- **High availability and monitoring**: RTO and RPO, hot/warm/cold sites, SNMP, syslog, NetFlow, baselines.
- **Command-line tools and what each shows**: `ping`, `tracert` / `traceroute`, `nslookup` / `dig`, `ipconfig` / `ifconfig` / `ip`, `arp`, `netstat`, `nmap`, packet capture, cable testers, tone generators, OTDR, spectrum analyser.

How your three labs map to the exam

Lab	Exam skills it proves
Home network	DHCP, static addressing, NAT, small network design, wireless basics
Small office	VLANs, trunking, inter-VLAN routing, subnetting, segmentation
Hotel network	Routing protocols (OSPF), WAN links, point-to-point subnets, SSH, device hardening

These labs give you the practical memory that makes scenario questions easier. They do not cover everything, so use the list above to fill the gaps: wireless design, cabling and connectors, DNS records, monitoring, disaster recovery and cloud.

A study method that works: Learn, Build, Recall, Apply

1. **Learn**: study one objective at a time from a video course or book, taking short notes in your own words. Always start from CompTIA's official objectives list.

2. **Build**: reproduce the concept in Packet Tracer or on real home equipment within a day. Concepts you have configured stay in memory.
3. **Recall**: close the notes and write the answer from memory. Use flashcards (Anki is free) with spaced repetition for ports, standards and acronyms.
4. **Apply**: do scenario questions on the topic. For every miss, write down why the right answer is right and why yours was wrong.

Tips for answering exam questions: read the whole scenario before the options; identify the OSI layer involved; eliminate two obviously wrong answers first; watch words like FIRST, BEST and MOST; answer PBQs on the second pass, after the easy questions are done.

Timeline for next year

Period	Focus	Goal
Now to Dec 2026	Finish Stage 1 and 2 of this guide, daily subnetting, ports flashcards	Subnet any /24 to /30 problem in under a minute
Jan to Feb 2027	Domains 1 and 2 (concepts and implementation) with labs	Pass chapter quizzes with 80% or higher
Mar to Apr 2027	Domains 3 and 4 (operations and security)	Explain each topic aloud without notes
May 2027	Domain 5 and the troubleshooting methodology, then full practice exams	Score 85% or more on three different practice exams
Book the exam	Only when you hit that score consistently	Sit the exam within about two weeks of booking

Adjust the months to your schedule. Five to eight hours a week is a realistic pace for most people studying alongside work or school.

Resources: CompTIA's official objectives PDF (free), Professor Messer's free N10-009 videos, a recent practice-exam set from a reputable provider, and Packet Tracer for labs. Avoid exam dumps. They are against CompTIA's rules, can get your certification revoked, and teach you nothing.

Sources: Global Knowledge, Network+ N10-009 overview, DestCert, N10-009 objectives explained. Exam details were checked on 12 October 2026 and can change, so confirm on CompTIA's website.